🌐 Day 2 Cheat Sheet: The Logical Map

1. Introduction to IP Addressing

An **IP (Internet Protocol) Address** is a numerical label assigned to each device connected to a computer network. While your MAC address (Day 1) is the "Serial Number" of the hardware, the IP is the **Logical Location** of the device.

📍 The Coordinate Analogy

Think of an IP address like **Latitude and Longitude**:

- **The Coordinates:** Just as $40.7128° N, 74.0060° W$ points to a specific spot on Earth, an IP points to a specific spot in a network.
- **The Logic:** You don't take your "Latitude" with you when you move; you get a new coordinate based on your new location. Similarly, when you join a new Wi-Fi, your IP changes to reflect your new "Neighborhood."

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2. The Hacker's Perspective

Why learn this? Because **Enumeration** (mapping the target) is the first step of any attack.

- **Targeting:** An IP is the unique "UID" of your victim.
- **The Perimeter:** Hackers look for "Public" vs "Private" IPs to see if they can attack from the internet or if they need to be inside the building.
- **The "Shout" (Broadcast):** Understanding the subnet tells a hacker exactly how many other computers they can "see" once they break into one machine.

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3. Deep Dive Technical: 32-Bit Hierarchy

IPv4 is a **32-bit binary string** divided into four 8-bit **octets**.

- **Structure:** 192 . 168 . 1 . 10
- **Binary View:** 11000000 . 10101000 . 00000001 . 00001010
- **Hierarchy:** * **Network ID (The Street):** The first portion of the bits.
  - **Host ID (The House):** The remaining bits.

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4. Subnetting (Fixed Length)

Subnetting is the act of drawing "walls" inside a network to manage traffic and security. We use a **Subnet Mask** to tell the computer where the "Street" ends and the "House" begins.

Class	Default Mask	CIDR	Purpose
Class A	$255.0.0.0$	/8	Massive (16 million hosts)
Class B	$255.255.0.0$	/16	Large (65,534 hosts)
Class C	$255.255.255.0$	/24	Small (254 hosts)

The Hacker Rule: If you are on a /24 network, you only need to scan **254 IPs**. If you are on a /16, you have **65,000+** potential targets to hide in.

[Image showing a single network block being divided into four equal subnets using a mask]

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5. IPv6: The Infinite Map

IPv4 is running out of "coordinates" (only 4.3 billion). IPv6 was built to replace it.

- **Size:** 128-bit (vs 32-bit).
- **Format:** Hexadecimal (e.g., 2001:0db8:85a3:0000:0000:8a2e:0370:7334).
- **The Shift:** * **No NAT:** Every device can have a "Global" IP.
  - **Auto-Config:** Devices can "assign themselves" an IP without a server.
- **Hacker Perspective:** Because IPv6 is "new" and "huge," many admins forget to secure it. A hacker can often bypass IPv4 firewalls by using the IPv6 "back door" that was left open by default.

💡 Day 2 Key Takeaway

"IPv4 is the map, Subnetting is the fence, and IPv6 is the expansion."

As a hacker, if you know the map and the fence, you know exactly how far you can run before you hit a wall.